

JEE MAIN REVISION MASTER: IUPAC NOMENCLATURES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	IUPAC Nomenclature & Functional Group Priority	5 PYQs	Nomenclature

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
IUPAC Nomenclature & Functional Group Priority	$\text{IUPAC Format: } \{\text{Prefix}\} + \{\text{Root Word}\} + \{\text{Primary Suffix}\} + \{\text{Secondary Suffix}\}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 IUPAC Nomenclature & Functional Group Priority (Moderate Yield — 5 PYQs)

Core Principle: IUPAC rules: Identify longest continuous carbon chain containing the principal functional group. Priority order of functional groups: $-\text{COOH} > -\text{SO}_3\text{H} > -\text{COOR} > -\text{COCl} > -\text{CONH}_2 > -\text{CN} > -\text{CHO} > >\text{C}=\text{O} > -\text{OH} > -\text{NH}_2 > >\text{C}=\text{C} > -\text{C}\equiv\text{C}-$.

■ **Caution / JEE Pitfall:** In numbering when double and triple bonds are at identical positions from opposite ends, double bond gets the lower locant ('en' before 'yne').

■ **Speed Trick:** Look at the principal suffix like -oic acid vs -oate vs -al to eliminate options quickly.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound with molecular formula $\text{C}_{20}\text{H}_{42}$ is..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The IUPAC name of the given organic compound is $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "IUPAC name of the compound is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound with molecular formula $C_{(20)}H_{(42)}$ is

- (A) Decane
- (B) Dodecane
- (C) Eicosane **[Correct]**
- (D) Hicosane

Correct Answer: C

Solution Analysis: The molecular formula is $C_{(20)}H_{(42)}$. For alkanes, the general formula is $C_{(n)}H_{(2n+2)}$. Here, $n=20$. So, $H=2(20)+2=42$. Hence the compound is an alkane with 20 carbon atoms. The alkane with 20 carbon atoms is called Eicosane. So, the correct option is Option C: Eicosane.